

Cost-Aware Evaluation of Multi-Agent LLM Systems for Thai Sarcasm Detection

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github.com/Thanaphumi2006/Cost-Aware-Multi-Agent-LLM-Evaluation-for-Thai-Sarcasm-Detection

ABSTRACT. Multi-agent LLM architectures — verifier pipelines, debate panels, hybrid committees — are widely assumed to justify their extra inference cost with higher quality. We test that assumption on a task where surface form and intent diverge sharply: sarcasm detection in Thai (ประชด). On a 127-item human-labeled gold set drawn from product reviews and tweets, we compare five systems under identical measurement code, tracking F1, API calls, dollar cost, and latency simultaneously. A two-agent screener-verifier pipeline initially appears to win (F1 0.744 vs. 0.690). The advantage does not survive a fair baseline: letting the single agent read its own token-level confidence and tuning one threshold — chosen leave-one-fold-out, at zero added cost — lifts it to 0.725, after which no multi-agent system separates from it (best Δ F1 +0.019, 95% CI [-0.073, +0.113], paired bootstrap). Model choice dominates architecture choice: a model 6 $\times$ cheaper than the flagship stays within a 0.05 F1 band, making a cheap single agent statistically indistinguishable from a flagship pipeline at one-eleventh the cost. We further establish a \$0 floor — three independent free methods (a 3-character regex, an embedding kNN, and a fine-tuned Thai transformer) all converge at F1 $\approx$ 0.59 — and show a three-way confidence router that retains 95% of flagship quality at 40% of the cost. Expanding the gold set 2.4 $\times$ (302 items, 67 positive) with adversarially plausible negatives narrows the pipeline-vs-tuned-baseline confidence interval threefold and preserves the null result, while collapsing precision for every architecture equally: hard negatives are a property of the data that additional agents do not repair. We conclude that for subjective, low-resource classification, the first questions are not architectural: exhaust the information one agent already returns, then choose the cheapest sufficient model.

1 Introduction

Multi-agent LLM systems promise quality through redundancy: multiple calls, multiple roles, a final adjudication. The promise is easy to believe and expensive to hold — every added agent multiplies API cost and latency. Whether the quality materializes is an empirical question, and one that is rarely asked with the cost axis in view.

We ask it on Thai sarcasm detection. Sarcasm (ประชด) is a deliberately adversarial target: the writer states the opposite of their intent, so surface cues mislead exactly the models that rely on them. Thai adds a low-resource setting, informal orthography (elongations such as ดีมากกก, the laughter numeral 555), and — in our source domains of restaurant reviews and tweets — a culture of exuberant sincere praise that is easy to mistake for irony. If multi-agent deliberation helps anywhere, a task defined by hidden intent should be favorable ground.

Our contributions:

- (1) A human-labeled Thai sarcasm gold set with a sharp operational rubric (sarcasm requires *feigned praise*, การเสแสร้ง), documented provenance and bias notes, plus a 2.4 $\times$ -expanded hard variant labeled blind through a purpose-built tool.
- (2) A five-system, four-axis comparison (F1, calls, cost, latency) under one measurement harness, with paired bootstrap and McNemar tests throughout.
- (3) A negative result with an identified mechanism: multi-agent gains over a *confidence-aware* single agent are statistically unresolvable while costing 1.8–7.4 $\times$, because half of the apparent gain was the baseline discarding confidence it had already paid for.

(4) Cost-axis anchors: a \$0 floor at $F1 \approx 0.59$ reached independently by three free methods, and a confidence router giving 95% of flagship quality at 40% of its cost.

2 Task and Data

Definition. An item is sarcastic (label 1) iff the author feigns praise or thanks while intending the opposite. The rubric resolves the three boundary cases that dominate disagreement: direct complaints are 0 (negative content without pretense), balanced reviews are 0 (genuine praise plus genuine criticism), and exuberant sincere praise is 0. Undecidable items (context-dependent intent) are excluded rather than guessed.

Gold set. 72,865 texts from Wongnai restaurant reviews and the Wisersight social-media corpus were keyword-scored for sarcasm suspicion; 400 sampled items were pre-labeled by GPT-4o and then human-reviewed *blind* (the annotator decides before seeing the model's draft). Agreement analysis showed the LLM over-flags sarcasm — of 17 LLM-positive items in the first review pass, the human kept 3 — motivating blind review as a design rule. Iterative mining of high-suspicion texts brought the canonical set to **127 items (30 sarcastic / 97 not)**. Provenance caveats (some positives were mined with the same model family later evaluated, likely inflating its recall) are documented and affect all systems equally.

Hard expansion. A later blind-labeling campaign over the remaining mined candidates added 175 items, giving **gold_v2: 302 items (67 sarcastic, 22%)**. Because candidates were pre-selected to *look* sarcastic and ~58% were judged sincere, gold_v2 is deliberately enriched with plausible negatives — a hard set whose absolute scores are not comparable to the canonical set (§6.4).

3 Systems

All LLM systems use the same detector prompt and JSON-constrained output on GPT-4o unless stated; prices are July-2026 list prices.

S1 Single agent. One call per item, argmax label. **S1+ τ Threshold.** Identical calls, but the positive-class probability is read from token logprobs and thresholded; τ is chosen leave-one-fold-out (stratified 5-fold, seed 42), so no item influences the threshold that judges it. **S2 Pipeline.** A screener flags candidates; a *constrained* verifier may only overturn 1→0 and only when confident, preserving the screener's recall. **S3 Debate.** Prosecutor and defender argue; a judge decides freely (3 calls/item). **S4 Hybrid.** A four-agent committee mixing the above roles. **S5 WangchanBERTa.** A Thai transformer fine-tuned on the gold set under 5-fold cross-validation $\times$ 3 seeds (every prediction from a model that never saw the item), run locally at \$0.

4 Evaluation Protocol

Accuracy is not reported: with 76% negatives, constant rejection scores 0.76. All systems predict the same items, so comparisons are paired: F1 differences carry 95% CIs from a paired bootstrap (5,000 resamples of the item set), and label flips are counted with McNemar's test. Costs are computed from measured token usage; latency is wall-clock per item (p50).

5 Results

System	F1	Prec.	Rec.	Calls	Cost	p50
S1 Single agent (argmax)	0.690	0.526	1.000	127	\$0.094	751 ms
S1+ τ Single agent + threshold	0.725	0.641	0.833	127	\$0.094	751 ms
S2 Two-agent pipeline	0.744	0.604	0.967	183	\$0.169	967 ms
S3 Debate (3 agents)	0.694	0.595	0.833	381	\$0.695	4,557 ms

System	F1	Prec.	Rec.	Calls	Cost	p50
S4 Hybrid (4 agents)	0.700	0.560	0.933	292	\$0.407	832 ms
S5 WangchanBERTa (local)	0.620	0.553	0.700	0	\$0.00	26 ms

TABLE 1. All systems on the same 127 items, same measurement code.

5.1 The fair baseline erases the multi-agent win

Against the plain single agent, the pipeline's +0.054 looks meaningful (95% CI $[-0.002, +0.113]$, $P(\text{not better}) = 3\%$). But that baseline throws away the model's own confidence. Reading it back — zero extra calls, zero extra dollars — lifts the single agent to 0.725, and against *that* baseline every multi-agent system is unresolved: pipeline +0.019 $[-0.073, +0.113]$ at $1.8\times$ cost; hybrid $-0.025 [-0.117, +0.071]$ at $4.3\times$; debate $-0.030 [-0.158, +0.096]$ at $7.4\times$. Roughly half of the apparent multi-agent gain was information the baseline had already paid for and discarded.

5.2 Architecture: constraints beat head-count

Why does the two-agent pipeline outperform the three-agent debate ($\Delta +0.049$, at a quarter of debate's price)? The detector starts from recall 1.000; what remains is buying precision without spending recall. The pipeline's verifier is *structurally constrained* — it can only overturn positive verdicts, and only when explicitly confident — so it keeps recall at 0.967 while pruning false positives. Debate returns full freedom to the panel: its defender successfully argues away 5 true positives (recall 0.833), spending the asset the detector had secured. The number of agents matters less than the constraint structure imposed on them.

5.3 Model choice dominates architecture choice

The tuned single agent across five OpenAI models spanning a $25\times$ price range stays within a 0.05 F1 band, with the best score from one of the cheapest (gpt-4.1-mini: F1 0.727, \$0.015). That cheap single agent is statistically tied with the flagship pipeline (gpt-4o, 0.744, \$0.169) at **1/11 of the cost**. Local open models (Qwen2.5-7B, SeaLLM-7B, Typhoon-8B, OpenThaiGPT-7B, 4-bit, chat-templated, same leave-fold-out protocol) top out at 0.576 — a real but modest gap of +0.152 to the GPT bot, and Thai-specialized models show no advantage over general Qwen.

5.4 Anchoring the cost axis

The \$0 floor. Three independent free methods — the regex 555 alone (F1 0.590), an e5 embedding kNN (0.588), and fine-tuned WangchanBERTa at the default threshold (0.590) — converge at ≈ 0.59 . The floor is a property of the task, not of any one method. **A useful middle.** A three-way router (confident-negative / uncertain $\rightarrow$ escalate to GPT / confident-positive) at a 40% escalation budget reaches F1 0.690 — 95% of the GPT bot — for 40% of its cost; beyond that the curve is flat. A cautionary side-result: tuning WangchanBERTa's threshold leave-fold-out *underperforms* not tuning at all (0.556 vs. 0.590) at this n — at small sample sizes, a tunable knob is a liability, and the 0.071 "gain" from tuning on the full set is leakage.

5.5 Replication at $2.4\times$ the data

System (gold_v2, n=302)	F1	Prec.	Rec.	Cost
S1 Single agent (argmax)	0.443	0.291	0.925	\$0.16
S1+ τ + threshold (LOFO)	0.458	0.325	0.776	\$0.16
S2 Two-agent pipeline	0.460	0.304	0.940	\$0.41 ($2.55\times$)

TABLE 2. Core systems on the expanded hard set. Absolute scores are not comparable to Table 1 by design (\$6.4).

On gold_v2 the pipeline-vs-baseline difference is +0.017 with 95% CI $[-0.013, +0.047]$ — an interval **three times tighter** than the original study's, still crossing zero, at $2.55\times$ the cost. The null result is not a small-sample artifact. Absolute F1 halves for every system because the added negatives are adversarially plausible (elongations, 555, effusive tone — sincerely meant): false positives surge identically across architectures. Hard negatives are a property of the data distribution that additional agents do not repair. The model is also overconfident here — 210/302 items receive $P(\text{sarcasm})$ outside $[0.01, 0.99]$ — leaving the threshold little probability mass to work with (even the leaked-optimal τ reaches only 0.492).

6 Limitations

(1) **Scale.** 30 positives (canonical) cannot resolve ~ 0.02 F1 differences; the expansion raises power but is not i.i.d. (2) **Provenance bias.** Some positives were mined with the model family under evaluation; the bias is shared across systems, so paired comparisons stand, but absolute recall should be read cautiously. (3) **Domain transfer.** On 55 hand-labeled Pantip forum comments, precision falls $0.68 \rightarrow 0.40$ (sincere praise flagged as sarcasm) while recall holds; deployment outside reviews/tweets requires re-tuning on target-domain labels. (4) **Single language/task.** Conclusions are demonstrated for Thai sarcasm; the *method* — fair confidence-aware baselines, paired statistics, cost as a first-class axis — is the transferable part.

7 Conclusion

On a task chosen to favor deliberation, multi-agent LLM systems did not separate from a single agent that simply reads its own confidence — while costing $1.8\text{--}7.4\times$ more. The result replicated at double the data with threefold-tighter intervals. Before adding agents, exhaust the one you have: read its confidence, tune with leakage-free protocol, and choose the cheapest sufficient model. Architecture is a question to ask after the information already purchased has been spent.

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Code, data, full statistics (findings 1–19), and a live demo: github.com/Thanaphumi2006/Cost-Aware-Multi-Agent-LLM-Evaluation-for-Thai-Sarcasm-Detection · Live detector: thanaphumi2006.github.io/Cost-Aware-Multi-Agent-LLM-Evaluation-for-Thai-Sarcasm-Detection/app.html